Граници на редици - задачи

1. Намерете границите на следните редици:

$$\lim_{n\to\infty}\left(\frac{3n^3+n-1}{3n^3-1}\right)^{n^2+2}= \tag{1}$$

$$\lim_{n\to\infty}\frac{5n^2+n^3.3^{2n-1}+5e^{2n}}{\operatorname{arctg} n^2+n.4^n+(n!)^{-2}}= \tag{2}$$

$$\lim_{n\to\infty}(n+1+n\cos n)^{(2n+n\cos n)^{-1}}= \tag{3}$$

$$\lim_{n\to\infty} n^3\left(\sqrt{n^2+\sqrt{n^4+1}}-n\sqrt{2}\right)= \tag{4}$$

$$\lim_{n\to\infty}\left(\frac{1+\sqrt[n]{a}}{2}\right)^n= \tag{5}$$

$$\lim_{n\to\infty}\frac{1}{2^n}\left(\sqrt[n]{\frac{n}{2n+1}}+\sqrt[n]{\frac{n^2}{2n^2+1}}\right)^n= \tag{6}$$

$$\lim_{n\to\infty}\frac{\lg\left(n^2+2n\cos n+1\right)}{1+\ln(n+1)}= \tag{7}$$

$$\lim_{n\to\infty}\sqrt[n]{\frac{2n^2-5n+3}{n^5+1}}= \tag{8}$$

$$\lim_{n\to\infty}\sqrt[n]{\frac{n^2+4^n}{n+5^n}}= \tag{9}$$

$$\lim_{n\to\infty}\frac{n-\lg n}{\log_2\left(4^n+1\right)}= \tag{10}$$

$$\lim_{n\to\infty}\left(\frac{n+1}{n+2}\right)^{\frac{1-\sqrt{n}}{1-n}}= \tag{11}$$

$$\lim_{n\to\infty}\left(\frac{3n^2-n+1}{2n^2+n+1}\right)^{\frac{n^3}{1-n}}= \tag{12}$$

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{\sqrt{n}}\operatorname{arctg}\left(\frac{n^4+3}{n^4+3\cos^3 n}\right)= \tag{13}$$

$$\lim_{n\to\infty}\sqrt[n^2]{\frac{n-1}{n+1}}+\sqrt[n]{3^n n^3+2}= \tag{14}$$

$$\lim_{n\to\infty} n\left(\frac{n+1}{2n-1}\right)^n\cos\left(\frac{n^4+3}{n^7+3\ln^3 n}\right)= \tag{15}$$

$$\lim_{n\to\infty}\frac{n\sqrt[n]{\lg n}}{\left(1+\frac{1}{n}\right)^{n^2}}= \tag{16}$$

$$\lim_{n\to\infty}\frac{\lg\left(5n^2+3n+1\right)}{\sqrt{n}+1}= \tag{17}$$

$$\lim_{n\to\infty}\frac{\operatorname{arccotg}\left(-n^2+3n+1\right)}{\arcsin\frac{n+1}{n+2}}= \tag{18}$$

$$\lim_{n\to\infty}\frac{\arccos\left(\frac{2n+3}{n^3+3}\right)}{(n^2-3)\ln\frac{n^2+1}{n^2+2}}= \tag{19}$$

2. Като се използва следния полезен факт:
(**Нека** $a_n > 0$ **и** $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = a$**. Тогава** $\lim_{n\to\infty} \sqrt[n]{a_n} = a$.)
да се намерят границите на следните редици:

$$\lim_{n\to\infty} \frac{1}{\sqrt[n]{n!}} = \tag{20}$$

$$\lim_{n\to\infty} \frac{n!}{(2n+1)!!} = \tag{21}$$

$$\lim_{n\to\infty} \frac{(2n-1)!!}{(2n)!!} = \tag{22}$$

$$\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n} = \tag{23}$$

$$\lim_{n\to\infty} \frac{4^n n!}{(3n)^n} = \tag{24}$$

$$\lim_{n\to\infty} \frac{3^n n!}{n^n} = \tag{25}$$

$$\lim_{n\to\infty} \sqrt[n]{(n+1)(n+2)...(2n)} = \tag{26}$$

3. Като се използва критерия на Коши, да се провери дали следните редици с общ член a_n са сходящи:

$$a_n = \sum_{k=1}^{n} \frac{1}{k^{3/2}} \tag{27}$$

$$a_n = \sum_{k=1}^{n} \frac{1}{k^2} \tag{28}$$

$$a_n = \sum_{k=1}^{n+1} \frac{\sin k}{2^k} \tag{29}$$

$$a_n = \sum_{k=2}^{n} \frac{1}{\ln k} \tag{30}$$

$$a_n = \sum_{k=1}^{n} \frac{\cos k!}{k(k+1)} \tag{31}$$

$$a_n = \sum_{k=1}^{n+1} \frac{1}{\sqrt{n^2+k}} \tag{32}$$

4. Да се покаже, че редицата с общ член

$$a_n = \sum_{k=1}^{n} \frac{1}{k} - \ln n$$

е сходяща. Границата и се бележи с $\gamma = 0.57...$ - константа на Ойлер.

5. Да се намерят границите на следните рекурентно зададени редици:

$$1) a_1 = a > -1, \quad a_{n+1} = \frac{1}{1+a_n} \tag{33}$$

$$2) a_1 = -7/13, \quad a_{n+1} = \frac{1+a_n}{2a_n} \tag{34}$$

$$3) a_1 = 3, \quad a_{n+1} = 1 + \frac{6}{a_n} \tag{35}$$

$$4) a_1 = 4, \quad a_{n+1} = \sqrt{6+a_n} \tag{36}$$